Anshuman Kumar

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Education

The City College of New York, CUNY **GPA 4.0**
Masters in computer science *Fall.2025 – Ongoing*

Vellore Institute of Technology, Vellore, India **CGPA. 8.91**
B.Tech in Computer Science with specialization in information security *Sep. 2021 – June 2025*

Experience

College Assistant (Tutor) Oct 2025 – Dec 2025
Grove School of Engineering (GSOE), CCNY | Fall 2025

- Tutored undergraduate students in core CS subjects including Algorithms, Databases, Software Engineering, Probability & Statistics and Discrete Mathematics.
- Provided 1:1 and group academic support, helping students strengthen fundamentals and improve exam performance.

WEB DEVELOPER Sep 2023 – Nov 2023
Tata Consultancy Services (TCS), Gurugram, India

- Collaborated with a team to manage a corporate website, focusing on optimizing performance, security, and user experience
- Developed prototype features using Vanilla frontend technologies like HTML, CSS, and JavaScript
- Gained experience with frontend frameworks such as Bootstrap and Tailwind, applying them to improve UI design and responsiveness.

Projects

RoboPety – AI Pet Simulation App (GCP + Docker + CI/CD + RBAC)

- Built and deployed an AI-driven pet simulation app using Flask, Docker, Cloud Run, achieving **99.5% uptime and 35% faster response times** through caching and container optimization.
- Implemented GitHub Actions CI/CD, **reducing deployment time by 70%** and automating security/vulnerability checks.
- Configured RBAC via IAM (Admin/Developer/Viewer roles), decreasing **security alerts by 40%** and ensuring least-privilege access with Two-factor Authentication (TOTP).
- Added Secret Manager, reCAPTCHA, and audit logging, **reducing unauthorized/bot traffic by 60%.**

AI Study Assistant – LLM-Powered Learning Tool (Python + HuggingFace + RBAC)

- Developed an AI study assistant using **LLMs (FLAN-T5/BERT)** to generate summaries and quizzes, improving output accuracy by **45%** and reducing hallucinations by **40%**.
- Integrated **RBAC** (Admin/User/Guest) for secure multi-level access, lowering unauthorized usage by **70%**.
- Optimized inference using quantization and batching, achieving **2× faster processing** and **35% lower GPU memory usage**.
- Added semantic search (SBERT embeddings) enabling **48% more accurate retrieval** from uploaded notes/PDFs.

Real-Time Voice Translator – Whisper + Deep Filtering Network (AI/ML Project)

- Built a real-time multilingual speech translation system using Whisper, fine-tuned on Common Voice, **improving translation accuracy by 42% and tone preservation**.
- **Applied Deep Filtering Networks (DFN) for noise suppression**, enhancing **speech clarity by 35%** across varied acoustic environments.
- Fine-tuned Whisper on a Hindi → English dataset, **reducing word-error rate (WER) by 28%** and improving inference consistency.

- Optimized model pipeline with batching and audio preprocessing, **achieving 1.8× faster real-time performance.**

Technical Skills

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, PySpark
Frameworks: React, Node.js, Flask
Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse
Project Management, AI Skill, Deep Neural Network
Cloud: GCP, AWS, AZURE

Certificates

- **Software Project Management (2022) (NPTEL)**
- **Google Cloud Practitioner Foundations (2023)**
- **Microsoft Certified: Power Platform App Maker Associate (PL-100) (2023)**